Woochang Sim

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Research Profile

Ph.D. researcher with published work in **controlled LLM reasoning evaluation** and experience developing **executable synthetic benchmarks**. My current research studies diagnostics for automated LLM-agent harness improvement and learned internal readouts for model judgment and self-monitoring. I connect these questions through algorithm implementation, controlled experiments, and analysis of model behavior; adaptive use of computation is a further research goal.

Broader interests: Adaptive agents under uncertainty and distribution shift; probabilistic task and environment modeling; Bayesian and meta-reinforcement learning; generalization to unseen tasks and environments; language-model priors and world models for interactive adaptation.

Education

Ph.D. in AI Convergence, Gwangju Institute of Science and Technology 2025–Present
Advisor: Prof. Sundong Kim

M.S. in Artificial Intelligence, Gwangju Institute of Science and Technology 2023–2025
Advisor: Prof. Sundong Kim

B.S. in Computer Engineering, Chosun University 2016–2023

Research Experience

Harness Diagnostics for Interactive LLM Agents *2026–Present*
Ongoing Ph.D. research; ARC-AGI-3 testbed

- Developed and piloted diagnostic tooling that exposes solver token probabilities and traces recorded context-production paths to support automated harness edits.
- Designed controlled comparisons that separate diagnostic judgments from executed behavioral outcomes and keep development tasks separate from final evaluation tasks.
- Analyzed recurring agent failures and incomplete provenance mappings; tool use and repair effects remain under pilot evaluation.

Internal Readouts for Model Judgment and Self-Monitoring *2026–Present*
Ongoing Ph.D. research

- Investigating learned readouts from unused output coordinates for model judgment while preserving language-generation ability.
- Studying whether these signals can track reasoning progress and inform self-checking and future compute allocation; signal validity, intervention effects, and allocation benefits are separate research questions.

Controlled Evaluation of LLM Reasoning on ARC *Nov 2023–Jan 2025*
Co-first author, ACM Transactions on Intelligent Systems and Technology (TIST)

- Operationalized a Language of Thought framework into reproducible experiments on **logical coherence, compositionality, and productivity**.
- Led implementation of controlled reasoning experiments, including prompting-based evaluations, an action-based compositionality environment, and an interface for human baselines.
- Compared language models with human performance and analyzed gaps between task accuracy and valid abstract and compositional reasoning.

GIFARC: Synthetic Analogy-Based ARC Tasks *Apr 2025–Present*
arXiv preprint; project

- Developed a **three-stage VLM/LLM generation pipeline** that converts GIF-derived analogies into visual abstractions, task sketches, and executable ARC-style task instances.
- Evaluated analogical information through controlled **in-context learning and supervised fine-tuning** experiments, including downstream ARC benchmarks.
- Coordinated experimental planning, task allocation, and weekly research discussions across collaborators.

Publications

∗ denotes equal contribution.

Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus
S. Lee*, **W. Sim***, D. Shin*, W. Seo, J. Park, S. Lee, S. Hwang, S. Kim, S. Kim.
ACM TIST, 2025. doi:10.1145/3712701; code.

GIFARC: Synthetic Dataset for Leveraging Human-Intuitive Analogies to Elevate AI Reasoning
W. Sim*, H. Ryu*, K. Choi*, S. Han, S. Kim.
arXiv preprint, 2025. arXiv:2505.20672.

Reasoning Abilities of Large Language Models through the Lens of Abstraction and Reasoning
S. Lee, **W. Sim**, D. Shin, S. Kim, S. Kim.
NeurIPS System-2 Reasoning at Scale **Workshop**, 2024. paper.

Enhancing Analogical Reasoning in the Abstraction and Reasoning Corpus via Model-Based RL
J. Lee*, **W. Sim***, S. Kim, S. Kim.
IJCAI IARML **Workshop**, 2024. arXiv:2408.14855.

The Possibility of Prompt Engineering for ARC Problem Solving
W. Sim, H. Jin, S. Kim, S. Kim.
Journal of KIISE: Computing Practices and Letters, 30(2):63–69, 2024. paper.

Using Contrastive Learning for Abstraction and Reasoning Task
G. Gu, **W. Sim**, J. Im, S. Kim, S. Kim.
Korea Software Congress, pp. 828–830, 2023. paper.

Augmenting Few-Shot Demonstrations with Large Language Model
W. Seo, **W. Sim**, S. Kim.
Korea Software Congress, pp. 341–343, 2023. paper.

Technical Skills

Programming: Python.
Frameworks and tools: PyTorch, Hugging Face Transformers, Weights & Biases, Git.
Research methods: Experimental design, benchmark construction, ablation studies, statistical evaluation, human evaluation, inter-annotator agreement analysis.
ML experience: LLM fine-tuning, in-context learning, synthetic data generation, information retrieval, open-domain question answering, data pipelines.
Languages: Korean (native), English (intermediate).

Selected Projects

Open-Domain Question Answering System [code] Oct–Nov 2021
Implemented a hybrid BM25/Elasticsearch/DPR retrieval pipeline and a RoBERTa-large reader over 60K Wikipedia documents; achieved **68.89 EM / 78.69 F1**, ranking 6th among 19 teams.

Relation Extraction with Typed Entity Markers [code] Sep–Oct 2021
Implemented typed entity markers and domain-adaptive masked-language-model pretraining with RoBERTa-large; achieved **72.6 F1**.

K-pop Relation Extraction Dataset Construction Nov 2021
Constructed a 2,468-instance dataset across 10 relation classes and developed iterative annotation guidelines; achieved Fleiss' $\kappa = 0.842$.

Teaching and Service

Teaching Assistant, Machine Learning & Deep Learning, GIST Sep–Dec 2025
Graded machine-learning assignments with detailed feedback and taught fundamental neural-network, convolutional-network, and recurrent-network concepts.

Student Representative, AI Convergence Department, GIST Jan–Dec 2025
Organized student events and facilitated communication between students and faculty.

Awards

The Chungshim Fellowship Mar 2024
Best Presentation Award, Korea Computer Congress 2023 Jul 2023